Hsuan-Te Lin (林軒德)

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Summary

Full-stack and full-cycle engineer with **eight years of experience** delivering web, mobile, backend, and cloud infrastructure solutions. Proven track record of taking products from greenfield development through production deployment and app store release, including a social platform built independently, a regulated fintech application delivered as primary developer, and an enterprise LMS serving **250,000+ registered accounts**. Combines hands-on product engineering with production ownership across **15 Kubernetes clusters** in multi-cloud and on-premises environments.

Technical Skills

- **Languages:** Python, JavaScript / TypeScript, Java (working knowledge), SQL, Bash / Shell
- **Backend / Data:** REST API design, relational data modeling, transactions, pagination, background jobs & scheduling, PostgreSQL (replication/failover, migrations, drift governance), MongoDB, Redis / Valkey, RabbitMQ / AMQP, OpenSearch, MinIO (S3-compatible), OAuth / JWT / SSO
- **Frontend / Mobile:** React, Next.js, React Native (iOS/Android), Redux, MUI, React Query, ECharts, i18n, CodePush OTA, App Store / Google Play / Enterprise distribution
- **CI/CD & Deployment:** GitLab CI/CD, ArgoCD / GitOps, Harbor, Docker, macOS build runners, App Store / Google Play release pipelines, Infrastructure as Code (AWS CDK; Terraform – familiar)
- **Containers / Cloud:** Kubernetes (EKS / AKS / Alibaba Cloud / on-premises), Helm, Rancher; AWS (EKS, CDK, Lambda, S3), Azure (Front Door, Blob Storage), Alibaba Cloud, Cloudflare, Vercel; Linux, Nginx, Traefik, CDN, DNS
- **AI / ML Systems:** Anomaly detection, SHAP explainability, LLM application workflows, multi-model and agent orchestration, MCP tool integration, RAG, knowledge graphs
- **Observability:** Prometheus, Grafana, ELK, Sentry
- **Security:** CIS Benchmark, OWASP / penetration-test remediation, Nessus, Cloudflare API Security, SonarQube, Fortify

Work Experience

Delta Electronics — Senior Software Engineer → Site Reliability Engineer

Mar 2020 – Present · Taiwan · Concurrent responsibility for product delivery and production platform operations

Application Development — Web & Mobile (2020 – 2025)

- Built and shipped an enterprise LMS app (**React Native**, iOS/Android) as one of its **two lead contributors — 1,650+ commits across 4 years** — on a **single-app, multi-tenant architecture** with config-server-driven environment and feature toggling
- Core contributor (**900+ commits**) to the **Next.js 14 web platform & admin console** (MUI / React Query / ECharts) — course-authoring studio, learner & instructor management, reporting dashboards, and the learner-facing site (explore, search, enrollment, commerce)
- Owned the end-to-end **app release pipeline** — App Store (consumer and enterprise) and Google Play, **CodePush OTA** updates, APNs/FCM push notifications, and the **macOS GitLab CI/CD build runners** behind them; ran SonarQube and Fortify scans and remediated critical- and high-severity findings

Backend Services

- Own several LMS backend services, including an **AWS Lambda media-processing pipeline** (primary author), a **WebSocket real-time hub**, and notification and AI-assistant services

- Diagnosed and fixed a top-priority **concurrent seat-contention race condition** that caused learners to lose seats mid-session at capacity; hardened study-record delivery with **AMQP**; implemented an identity **account lifecycle** supporting soft deletion and reactivation while preserving UID and audit history
- Solo-authored a **Python anomaly-detection and root-cause pipeline** for *ACES*, an award-winning data-center monitoring project with Delta Research Center — telemetry → anomaly detection → per-row **SHAP** attribution → **LLM root-cause analysis**, with results written back over REST; also built the project's **3D visualization demo** (React Three Fiber)
- Contribute across services written primarily in **TypeScript/Node** and **Python**, and delivered **Java** features after ramping up on an unfamiliar production codebase

CI/CD, Cloud Infrastructure & Security (2022 – Present)

- Operate **15 Kubernetes clusters** (AWS EKS, Azure AKS, Alibaba Cloud, and on-premises) running the LMS platform across five environments — **250,000+ registered accounts and 3,000 peak concurrent users**, with one multi-tenant cluster hosting **15+ business units and brands**; **cut compute costs by approximately 40%** through an AWS Spot migration
- Own the deployment path end-to-end: **GitLab CI/CD pipelines, ArgoCD/GitOps, Dockerfiles, and Harbor registry**; completed **zero-downtime cluster upgrades spanning 10 Kubernetes versions**; led production incident response involving etcd/control-plane recovery and CoreDNS
- **Run the data layer** — PostgreSQL, MongoDB, Redis, RabbitMQ, OpenSearch, and MinIO — with per-tenant isolation and replicated PostgreSQL with automated failover; own schema migrations and production-to-development data synchronization
- **Bootstrapped a greenfield Kubernetes platform** for Delta Electronics Thailand's new EV application — replicated PostgreSQL, Traefik ingress, policy-as-code, and observability from scratch; defined the Git branching conventions & engineering workflows the in-country team now builds on
- Remediated a brute-force login vulnerability identified through penetration testing across the backend, frontend, and infrastructure; hardened IIS against the **CIS Benchmark**; ran Nessus scans; strengthened Cloudflare API security controls

Aurora Technology — Frontend Engineer

Oct 2018 – Mar 2020

- Built web frontends in **React** and cross-platform mobile apps in **React Native** (iOS/Android)

Earlier — Research (2015 – 2017)

- Built drought / water-supply-risk early-warning and decision-support systems (Formosa Plastics Group; UMC); thesis graded A+, research presented at the European Geosciences Union (EGU) 2016

Selected Projects

Social Platform (web) — designed and built solo · TypeScript

- Took it from empty repo to deployed product: **Next.js 15 / React 19** over **193 API route handlers**, a **67-module service layer**, and a **PostgreSQL schema of 87 tables / 161 relations / 113 indexes** via Prisma
- Owned the backend end-to-end — **Zod validation at every request boundary**, multi-provider **OAuth + JWT**, transactional writes with cursor pagination, rate limiting, **scheduled jobs**, **SSE** realtime, and a two-stage content-moderation pipeline; gated on **GitHub Actions CI** with Vitest and Playwright

Consumer Lending App (iOS / Android) — contract engagement, primary developer

- Took a regulated fintech client app from zero to **store release** — **React Native / TypeScript** with MobX-State-Tree, **biometric login**, camera-based **KYC** document capture, loan and repayment flows integrated against the client's APIs, three-language i18n, and Detox E2E tests

AI Multi-Agent Ops Platform — personal project · Python

- Built a **six-agent operations platform** that coordinates **monitoring, incident response, and development** through a unified Kanban workflow; added human-in-the-loop safeguards that restrict agents to branch-only commits with no push or deployment access, and validated its read-only integrations against a live Kubernetes cluster

Awards

- **Honorable Mention, Enterprise Division — Ministry of Economic Affairs "AI Smart Innovation Award" (AI 智能創新大賞)**, with Delta Research Center (project *ACES*)

Education & Languages

M.S. (2017) / **B.S.** (2015), Bio-Environmental Systems Engineering, National Taiwan University · Chinese (native), English (professional), Japanese (basic)